Seminar 1. (1.10.2025)

Operation: $* : A \times A \to A$ ⤳ parte stabilă ⤳ stable subset

$(\forall)\, x, y \in A \Rightarrow x * y \in A$

Semigroup: $(A, *)$ ⤳ asoc.

$* \to$ ass: $(\forall)\, x, y, z \in A,\ (x*y)*z = x*(y*z)$

Monoid: $(A, *)$ semigroup ⤳ asoc. + e.n.

$* \to$ has an $\underbrace{\text{identity element}}$

$\exists e \in A, (\forall) x \in A$ st. $x * e = e * x = x$

Group: $(A, *)$ monoid ⤳ asoc + e.n. + inv.

+ each element $\underbrace{\text{has an invers,}}$

$(\forall) x \in A^*, \exists x^{-1} \in A^*$ s.t. $x * x^{-1} = x^{-1} * x = e$

Abelian group / Commutative group $\begin{cases} (A, *) \text{ group} \\ * \to \underbrace{\text{comm.}} \end{cases}$ ⤳ asoc + com + e.n. + inv.

$(\forall) x, y \in A : x * y = y * x$

Subgroup: $(G, *)$ group, $H \subseteq G$
then H is a subgroup of G ($H \leq G$) if
H is a stable subset (part) of G and
$(H, *)$ is also a group.

① $+, -, \cdot, :$
On $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$
are they operations?

$(\mathbb{N}, +)$ ✓
$(\mathbb{N}, -)$ ✗ → ctr. $(1-2 \notin \mathbb{N})$
$(\mathbb{N}, \cdot)$ ✓
$(\mathbb{N}, :)$ ✗ $(2 : 3 \notin \mathbb{N})$

$(\mathbb{Z}, +)$ ✓
$(\mathbb{Z}, -)$ ✓
$(\mathbb{Z}, \cdot)$ ✓
$(\mathbb{Z}, :)$ ✗ $(-5 : 3)$
$(\mathbb{Z}^*, :)$ ✗
$(\mathbb{N}^*, :)$ ✗

$(\mathbb{Q}, +)$ ✓
$(\mathbb{Q}, -)$ ✓
$(\mathbb{Q}, \cdot)$ ✓
$(\mathbb{Q}, :)$ ✗ $(7 : 0 \,\nexists)$
$(\mathbb{Q}^*, :)$ ✓
$(\mathbb{R}^*, :)$ ✓
$(\mathbb{C}^*, :)$ ✓

$+, -, \cdot$ for $\mathbb{R}, \mathbb{C}$
$:$ only for $\mathbb{R}^*, \mathbb{C}^*$

③ $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$

$+, \cdot$

are they groups?

- ✗ $(\mathbb{N}, +)$: as. ✓, id ✓, inv. ✗
- ✓ $(\mathbb{Z}, +)$: as. ✓, id. ✓, inv. ✓
- ✗ $(\mathbb{N}, \cdot)$: as. ✓, id. ✓, inv. ✗
- ✗ $(\mathbb{Z}, \cdot)$: ✓, ✓, ✗

$(\mathbb{Q}, +), (\mathbb{R}, +), (\mathbb{C}, +)$ ✓

$(\mathbb{Q}^*, \cdot), (\mathbb{R}^*, \cdot), (\mathbb{C}^*, \cdot)$ ✓

$(\mathbb{Q}, \cdot), (\mathbb{R}, \cdot), (\mathbb{C}, \cdot)$ ✗

④ $(\mathbb{R}, *)$, $x * y = x + y + xy$, $(\forall)\, x, y \in \mathbb{R}$

a) $(\mathbb{R}, *)$ commutative monoid?

b) $[-1, \infty)$ stable subset of $(\mathbb{R}, *)$

Sol

a) Let $x, y \in \mathbb{R}$

$$\left.\begin{aligned} x * y &= x + y + xy \\ y * x &= y + x + yx \end{aligned}\right\} \Rightarrow (\mathbb{R}, *) \text{ comm } (1)$$

Let $x, y, z \in \mathbb{R}$ my freaking embarassing mistake

$$(x*y)*z = x + y + xy + z + (x + y + xy)\cdot z = xyz + xy + xz + yz + x + y + z$$
$$x*(y*z) = x + y + z + yz + x\cdot(y + z + yz) = xyz + xy + xz + yz + x + y + z$$

$\Rightarrow (\mathbb{R}, *)$ ass. (2)

$\exists\, e \in \mathbb{R}$ s.t. $x * e = e * x = x$, $(\forall)\, x \in \mathbb{R}$.

$$x * e = x + e + xe = x$$
$$\Leftrightarrow x(e+1) + e = x$$
$$\Leftrightarrow x \cdot e + e = 0$$
$$\Leftrightarrow e(x+1) = 0 \Rightarrow e = 0 \Rightarrow (\mathbb{R}, *) \text{ has id. elem. } (3)$$
$$\cancel{x = -1}$$

$(1), (2), (3) \Rightarrow (\mathbb{R}, *)$ comm. monoid.

b) $x, y \in [-1, \infty)$

$$x * y = x + y + xy = x(y+1) + y + 1 - 1 = (y+1)(x+1) - 1$$

$$x \geq -1 \Rightarrow x + 1 \geq 0$$
$$y \geq -1 \Rightarrow y + 1 \geq 0$$

$\Rightarrow (y+1)(x+1) \geq 0 \mid -1$

$\Rightarrow x * y \geq -1$

$\Rightarrow [-1, \infty)$ stable subset of $(\mathbb{R}, *)$

⑤ $(\mathbb{N}, *)$

$(\forall)\, x, y \in \mathbb{N} : \quad x * y = \gcd(x, y)$

a) $(\mathbb{N}, *)$ comm. monoid?

b) $D_m = \{x \in \mathbb{N} \mid x \mid m\}$, $m \in \mathbb{N}^*$

$(D_m, *)$ stable subset of $(\mathbb{N}, *)$?

$(D_m, *)$ comm. monoid?

c) Fill in the table of $(D_6, *)$

Sol.

a) $x * y = y * x$

$x * y = \gcd(x, y) = \gcd(y, x) = y * x, (\forall)\, x, y \in \mathbb{N} \Rightarrow (\mathbb{N}, *)$ comm. (1)

Specific enough for z?

★ Prove that $\gcd(\gcd(x, y), z) = \gcd(x, y, z)$

$\gcd(x, y) = t$

$t \mid x$ (1)

$t \mid y$ (2)

$\gcd(\gcd(x, y), z) = \gcd(t, z) = g$

$g \mid t \xrightarrow[(2)]{(1)} g \mid x;\ g \mid y \Rightarrow g = \gcd(x, y)$

$g \mid z$

$\left.\begin{matrix} g \mid t \\ t \mid x \end{matrix}\right\} \Rightarrow g \mid x$

$\Rightarrow$ ★ is true.

$\exists\, e \in \mathbb{N}$ s.t. $e * x = x * e = x$ ($\xleftarrow{\text{comm.}}$) $\gcd(x, e) = x \Rightarrow$

$\Rightarrow e = 0 \in \mathbb{N} \Rightarrow (\mathbb{N}, *)$ has id. $e = 0$. (2)

$(2, 0) = 2$

$(1, 8) = 1$

$(x * y) * z = x * (y * z)$

$\Leftrightarrow \gcd(\gcd(x, y), z) = \gcd(x, \gcd(y, z))$

$= \gcd(x, y, z)$ (★)

$\Rightarrow (\mathbb{N}, *)$ ass. (3)

(1), (2), (3) $\Rightarrow (\mathbb{N}, *)$ comm. monoid

b) • Let $x, y \in D_m$

?

$z = x * y = \gcd(x,y) \in D_m$

$$\left.\begin{array}{l} x, y \in D_m \Rightarrow x \mid m \\ \qquad\qquad\quad\; y \mid m \\ \gcd(x,y) = z \mid x \\ \gcd(x,y) = z \mid y \end{array}\right\} \Rightarrow z \mid m \Leftrightarrow x * y \in D_m \Rightarrow (D_m, *) \text{ stable subset of } (\mathbb{N}, *).$$

- Comm. – inherited from $(\mathbb{N}, *)$
  Ass. – inherited

$$\left.\begin{array}{l} \text{Comm. – inherited from } (\mathbb{N},*) \\ \text{Ass. – inherited} \\ (D_m, *) \text{ stable subset} \end{array}\right\} \Rightarrow (D_m, *) \begin{array}{l} \text{comm (1)} \\ \text{ass. (2)} \end{array}$$

$\exists e \in \mathbb{N}$ s.t. $e * x = x * e = x$ $\left(\xleftarrow{\text{comm.}}\right)$

$\Leftrightarrow x * e = x$

$\Leftrightarrow \gcd(x, e) = x$

$\Rightarrow e = m \in D_m$ (3)

(1),(2),(3) $\Rightarrow (D_m, *)$ comm. monoid.

d)

$*$	1	2	3	6
1	1	1	1	1
2	1	2	1	2
3	1	1	3	3
6	1	2	3	6

6) Determine the finite stable subsets of $(\mathbb{Z}, \cdot)$

Suppose $\left.\begin{array}{l} H \subseteq \mathbb{Z} \\ H \neq \emptyset \end{array}\right\} \Rightarrow \exists x \in H \Rightarrow \exists x \cdot x = x^2 \in H$

$\Rightarrow \exists x \cdot x \cdot x = x^3 \in H$

$\Rightarrow \exists x \cdot x \cdot \ldots \cdot x = x^n \in H$ $\xrightarrow{\text{finite}}$

If $|x| \geq 2 \to \infty$

$\Rightarrow \exists i, j \in \mathbb{N}^*$

$\left.\begin{array}{l} 0 < i < j \text{ s.t. } x^i = x^j \\ H \subseteq \mathbb{Z} \end{array}\right\} \Rightarrow \{-1, 0, 1\}$ → the only nr. that multiplied by themselves don't $\not\to \infty$

$\Rightarrow \{0\}, \{1\}, \{0,1\}, \{-1,1\}, \{-1,0,1\}$

but no $\{-1, 0\}$ because $(-1)\cdot(-1) = 1 \notin \{-1, 0\}$.

but $H = \{-1, 0\}$ ...

Explanation: $H=\{0\}$, $H=\{1\}$ ✓
$H=\{0,1\} \to 0\cdot 0 = 0 \in H$; $1\cdot 1 = 0 \in H$, $0\cdot 1 = 0 \in H$, $1\cdot 0 = 1 \in H$ ✓
but $H=\{-1,0\} \to (-1)\cdot(-1) = 1 \notin H$, $(-1)\cdot 0 = 1 \in H$ ... ✗

⑦ $(G,\cdot)$ group.

a) $(G,\cdot)$ is abelian $\Leftrightarrow (\forall)\, x,y \in G: (xy)^2 = x^2y^2$

* b) $x^2 = 1$ (e.m.), $(\forall)\, x \in G \Rightarrow G$ abelian.

Sol.

a) Hint: $(xy)^2 = xy \cdot xy$

b) Hint: $x^2 = 1 \Leftrightarrow x \cdot x = 1$
$\Leftrightarrow x = x^{-1}$.

b)
method 1

$(G,\cdot)$ group $\to$ ass. ✓
$\to$ identity elem ✓
$\to$ inverse ✓ $\to$ inv. of x is $x^{-1} = \frac{1}{x}$

$x^2 = 1 \Leftrightarrow x \cdot x = 1 \;\Big|\cdot x^{-1}$

$\Leftrightarrow x = x^{-1}$

$(\forall)\, x \in G$ has an inverse $\Rightarrow x \cdot x^{-1} = x \cdot \frac{1}{x} = 1 = x^{-1} \cdot x \Rightarrow$

$\Rightarrow (G,\cdot)$ abelian group.

Method 2

$x^2 = 1 \Leftrightarrow x \cdot x = 1 \,\Big|\cdot x^{-1}$, $(\forall)\, x \in G$

$\Longrightarrow x^{-1} = x$, $(\forall)\, x \in G$.

Let $x, y \in G$ arbitrary

$xy = x^{-1} \cdot y^{-1} = (xy)^{-1} \overset{(*)}{=} y^{-1} \cdot x^{-1}$.

(*) $(M,\cdot)$ monoid.
$x_1, x_2, \dots, x_n$ inv.
$\Downarrow$
$(x_1 \cdot x_2 \cdot \ldots \cdot x_n)$ inv.,
and
$(x_1 \cdot \ldots \cdot x_m)^{-1} = x_m^{-1} \cdot \ldots \cdot x_1^{-1}$

7 a) $(G,\cdot)$ is abelian $\Leftrightarrow (\forall)\, x,y \in G: (xy)^2 = x^2y^2$

„ $\Longrightarrow$" $(G,\cdot)$ abelian group
Let $x, y \in G$ $\Bigg\} \Rightarrow x \cdot y = y \cdot x, (\forall) x, y \in G.$

A $\underline{\text{reguli de calcul cu}}$
puteri într-un grup $\Rightarrow x^m \cdot y^n = y^n \cdot x^m, (\forall) n, m \in \mathbb{Z}.$

if $n = m = 2$

$\Rightarrow x^2 \cdot y^2 = y^2 \cdot x^2$

$(G,\cdot)$ abelian group $\Rightarrow xy = yx \in G.$

$xy \cdot xy \in G$